AUTOMATIC IRRIGATION SYSTEM USING IC 555 AND LM741

Analog Group Project

Submitted in partial fulfillment of the requirements of
ECE/EEE/INSTR F341 Analog Electronics

By
Group no 59

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CERTIFICATE

This is to certify that Group No. 59 with the following group members

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Have submitted the project report in partial fulfillment of the course Analog Electronics (ECE/EEE/INSTR F341) during the academic year 2020-21 for the Analog Electronics course requirement.

Date. **03.04.2021** ***Instructor's Signature***

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Moreover, we are also thankful to the institute for enabling us to take up this course and pursue this project.

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Introduction

The objective of the project is to build a plant irrigation setup. The unit can regulate the water level in the supply tanks and maintain the soil's required moisture contents. This was implemented using LM741 and IC 555, covered in the Analog Electronics Course. This also served as an opportunity to understand these components and other components used in greater detail.

Applications:

- It can be used in to control the water level. The circuit's output is in 12 V range, so the pumps having specifications similar to the range mentioned above need to be used accordingly. With adequate scaling, it can also be used for household purposes and farming purposes (irrigation).
- In many places, it is not possible to manually operate the pump. The water level controller makes this process automatic.
- The soil moisture testing unit can be used to maintain the moisture content and water the plants when required.

Advantages:

- Low maintenance.
- Compact and Fully Automatic design.
- It saves water, energy.
- It can be operated with the help of a 12V battery or adaptors (converts 230V AC to 12V DC) readily available in markets.
- It can be easily installed in water tanks because of the fewer components used in circuits.
- It helps in maintaining the minimum and maximum water level required in the water tank.
- With the simple IC 555 timer circuit, we can prevent water overflow from the water tank and thus save water. The pump will automatically start if the water level falls to the minimum level. So, there is no need to start and stop the pump manually.
- The complete unit makes the irrigation of plants much efficient as the water in the supply tank is monitored automatically, and similarly, soil moisture cannot drop below a permissible limit

List of Components

Components Name	Quantity
NE 555 Timer IC	x1
LM-741 Operational Amplifier	x1
BC547B NPN Transistor	x2
180k 0.25-watt Resistor	x1
22k 0.25-watt Resistor	x2
1k 0.25-watt Resistor	x2
1M 0.25-watt Resistor	x2
100k 0.25-watt Resistor	x1
10k 0.25-watt Resistor	x1
2.2k 0.25-watt Resistor	x1
100k Potentiometer	x1
10nF Capacitor	x1
1N4007 diode	x2
LED 1.5V 5-mm	x2
12 V SPDT Relay	x2
12V – Power Supply	x1

Block Diagram

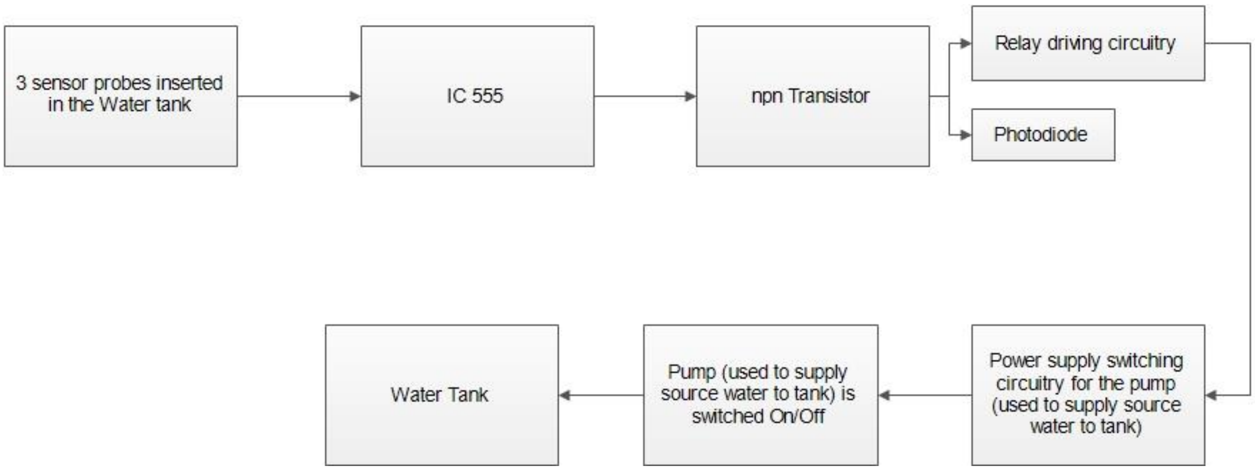


Figure 1: Water level monitoring system

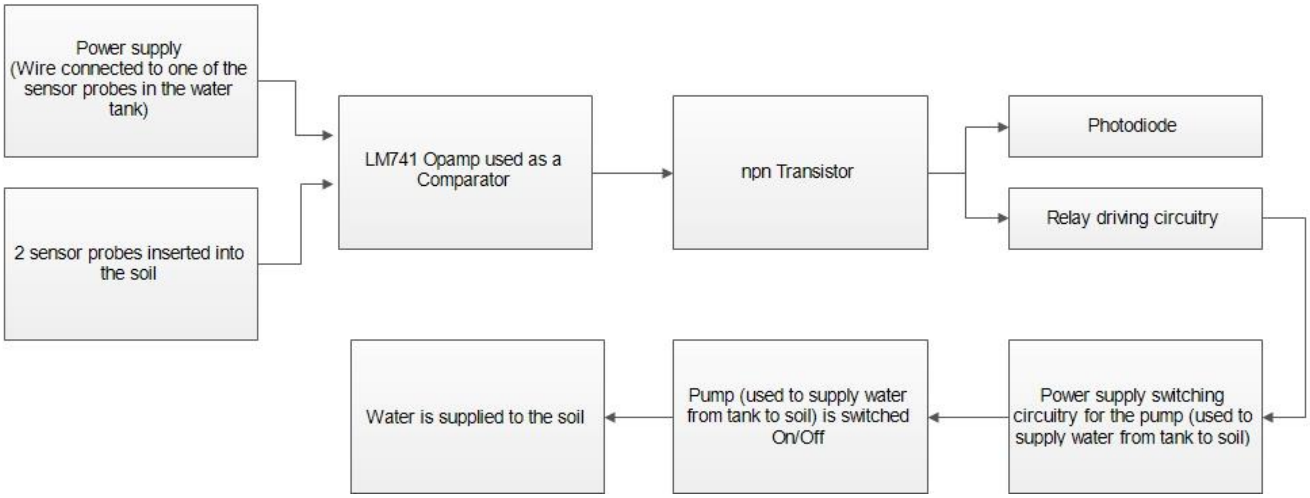


Figure 2: Soil Moisture Testing Unit

Circuit Diagram

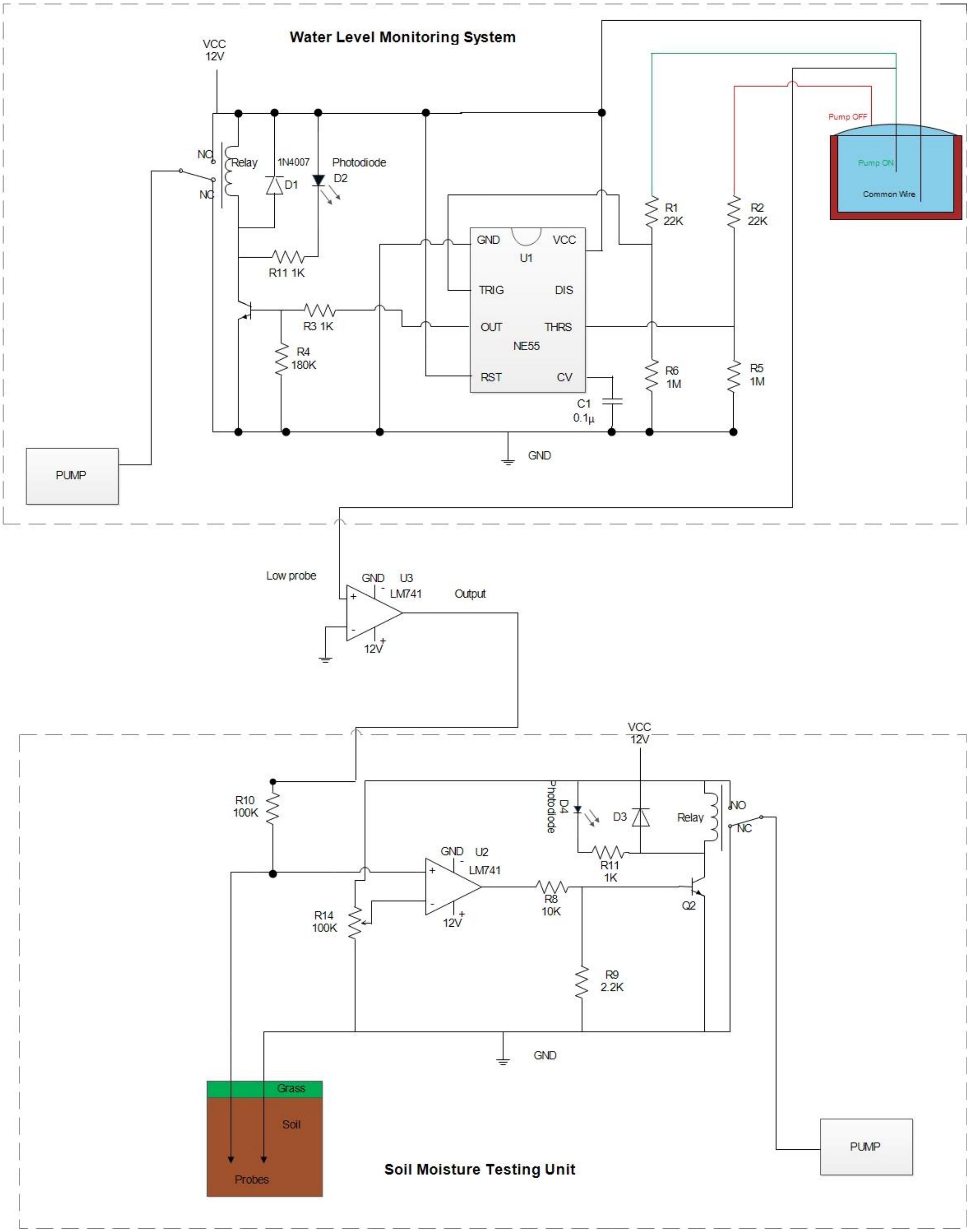


Figure 3: Circuit Diagram: Automatic Irrigation Setup

Functioning of the circuit

Automatic Water Level Monitoring system:

The circuit developed for this block is entirely automatic and is used to monitor and regulate the Water Level of the Tank, which acts as a water source for the irrigation setup. This circuit uses one NE 555 Timer IC, BC547 Transistor, SPDT Relay, and other passive components. There are three probes in the circuit which sense the level of water. The circuit uses two probes that are used to detect the level of water at respective positions in the tank. The third probe, 'common wire,' is connected to the 12V power supply.

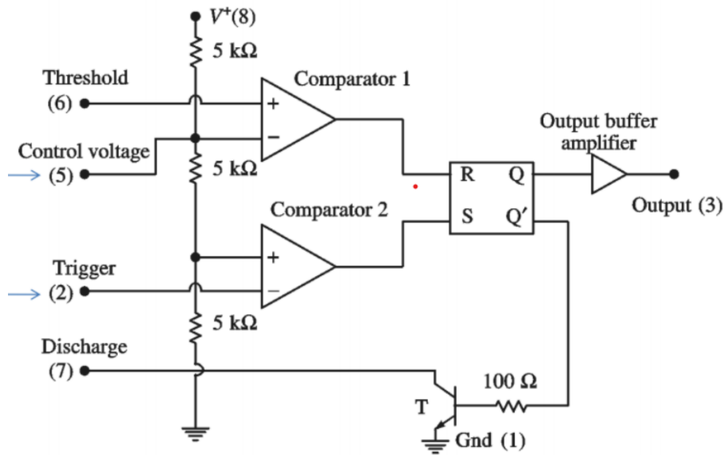


Figure 4: Block Schematic of IC555

For an IC 555, there are 2 comparators in its internal structure, as shown in the above figure. For Comparator 1, the value V_{TH} (Pin 5) is internally fixed to $\frac{2Vcc}{3}$. Since we don't intend to override, the comparator input is connected to bypass to the ground using a 10nF capacitance. The value of V_{TL} in Comparator 2 is fixed to $\frac{Vcc}{3}$.

The Trigger Input (Pin 2) is connected to the voltage divider between 'Low Probe" and the common ground. Similarly, the Threshold Input (Pin-6) is connected to the voltage divider between "High Probe" and the common ground. The value of resistors is chosen such that under a given circumstance, the input to Pin-2 and Pin-6 are above the $\frac{Vcc}{3}$ and $\frac{2Vcc}{3}$ respectively.

- When the water level is below the low probe. There is no contact between the 'Low Probe,' 'High Probe,' and the 'common wire.' Due to which the drop across the Trigger Pin and the Threshold Pin is zero, driving the comparator outputs' low' and 'high' respectively, i.e., R = 0; S = 1. This makes the output of the Timer IC high, which is further used to drive the relay in ON state.

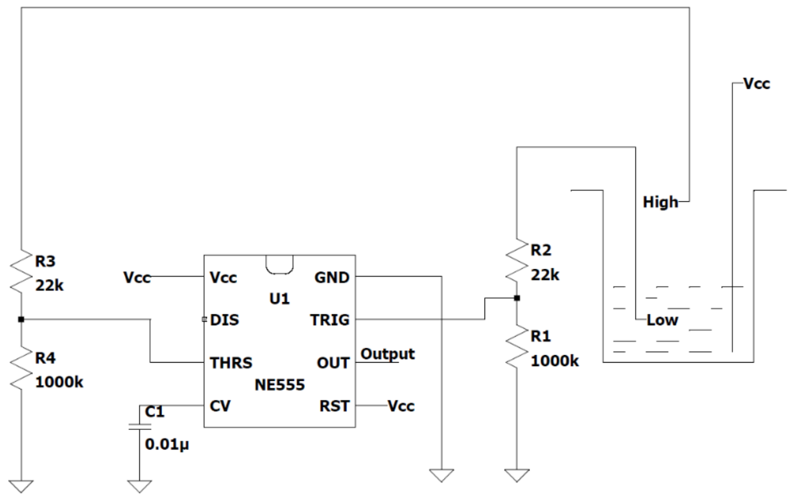


Figure 5: State When water Level is above Low probe

- When the water makes contact with the 'Low Probe,' the voltage drop across the Trigger Pin is greater than $\frac{Vcc}{3}$. Therefore, the comparator outputs are both low, i.e., R = 0, S = 0. Due to which the output of the RS flip-flop retains its previous state, and the output of IC 555 stays high, and the relay continues to remain in the ON state.

- In the last case, when the tank is full, water makes contact with the 'High Probe,' the voltage at pin 6 goes high and goes above the V_{TH} value on the inverting input of the Opamp, which gives the output of Comparator 1 as high, but the output of Comparator 2 is low. This sets the internal RS flip-flop of IC, and the output goes low. This low output at pin 3 switches off the transistor which de-energizes the relay and disconnects the pump power supply.

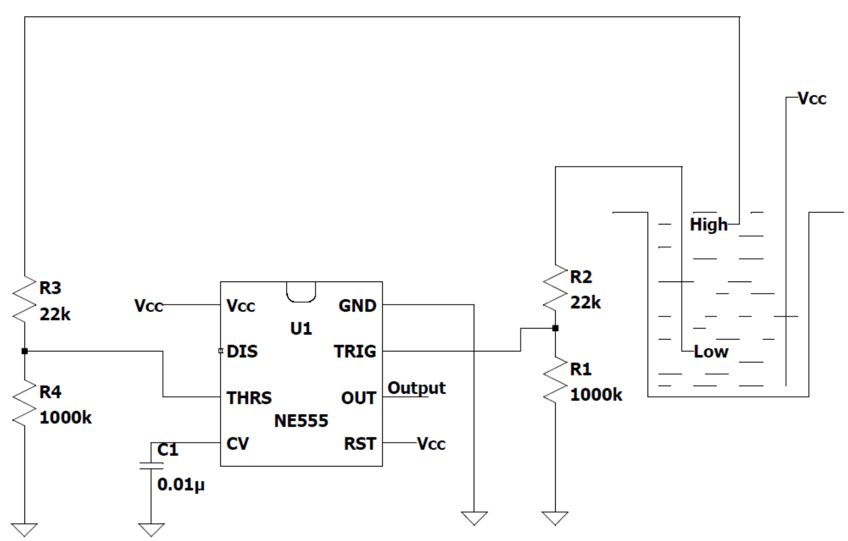


Figure 6: State When water Level is above High probe

Soil Moisture Testing Unit:

This circuit is used to monitor the soil's moisture level and supply water to it from the tank to meet the required moisture content of the soil. The circuit consists of one LM741 Opamp, BC547 transistor, relay, and required passive components. Two probes that are inserted into the soil are used for testing moisture levels. The soil resistance depends on the fact whether the soil is wet or dry. When the soil is wet, the resistance is low; hence the soil conducts adequately, while in the case when the soil is dry, its resistance increases, and the soil no longer conducts.

The potentiometer is used to calibrate the moisture content of the soil. This value of the potentiometer is used to set the reference voltage for the comparator used.

Working of the Comparator (U2):

1. When the soil moisture content is high, it produces lower resistance. As a result, the voltage divider circuit's output, i.e., the input voltage at the non-inverting terminal, is lower than the reference voltage that results in –Vsat value as the output of the comparator. In this case –Vsat equals low voltage (approximately 0). Thus, the output of the comparator is low, indicating that the soil is wet.
    a. For example, if the soil moisture content evaluates to 10K Ohm of resistance, then the value of the voltage at non-inverting input equal to $\frac{10K}{10K+100K} \times V_{sat}$ while the voltage in the inverting terminal is equal to $\frac{50K}{50K+50K} \times V_{sat}$, which is higher than the value at the non-inverting terminal, thus giving a low output.

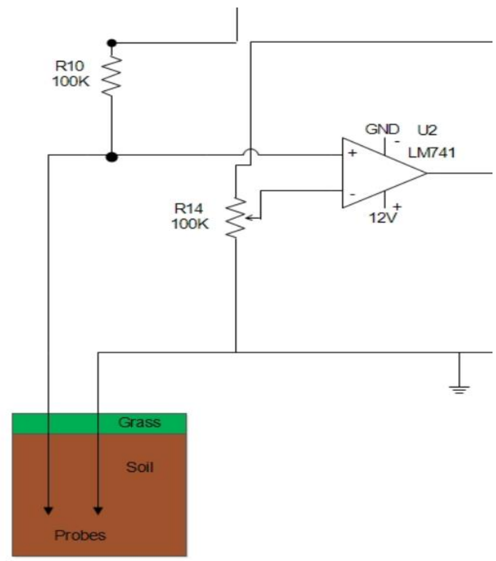


Fig 7. Soil Testing Unit

2. When the soil moisture content is low, it produces higher resistance. As a result, the voltage divider circuit's output, i.e., the input voltage at the non-inverting terminal, is higher than the reference voltage that results in +Vsat value at the output of the comparator. In this case, +Vsat equals high voltage (approximately 12V). Thus, the output of the comparator is high, indicating that the soil is dry.
    a. For example, if the soil moisture content evaluates to 200K Ohm of resistance, then the value of the voltage at non-inverting input equal to $\frac{200K}{200K+100K} \times 12$ while the voltage in the inverting terminal is equal to $\frac{50K}{50K+50K} \times 12$, which is lower than the value at the non-inverting terminal, thus giving a high output.

Working of the Comparator (U3):

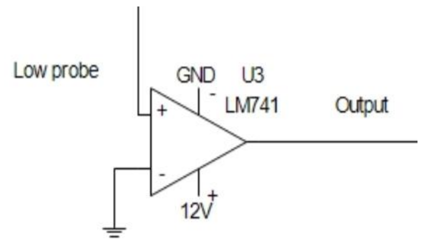


Fig 8.water level controlled switch

Additionally, a water level-controlled switch has been added to connect the Water Level Monitoring Block and Moisture Level Tester. This ensures the irrigation setup is turned on only when there is sufficient water in the tank. This is achieved by using an Operational Amplifier as Comparator Switch. The Non-Inverting terminal of Opamp is connected to the low probe while the non-inverting terminal is grounded. When the water level is above the low probe, the non-inverting terminal of Opamp is high. This output high voltage powers the moisture tester circuit.

Relay switch circuit

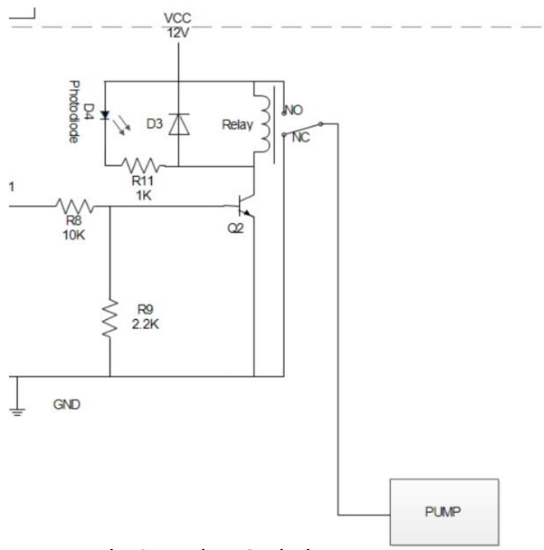


Fig 9. Relay Switch

The relay switch circuit has a coil driven by an NPN transistor. When the transistor's base voltage is zero, the transistor is in cut-off and does not allow the current to flow through it, thus not allowing the relay coil to charge up as they are in series. But when the large enough current is driven through the base of the transistor, the current starts to flow through the relay coil.

Now the relay consists of 3 pins, namely Common pin (CO), Normally Closed pin (NC), and Normally Open pin (NO). Initially, when the coil is demagnetized, the CO pin is connected to the NC pin. But when the coil gets magnetized, it shifts to NO pin.

The diode 1N4007 that is used in parallel to the relay coil is an essential component. When the transistor switches off after being on for some time, it stops the current through it. But the coil is still magnetized, and it requires a path to demagnetize itself; The diode acts as a freewheeling diode and allows the coil to demagnetize itself, thus protecting the circuit from any damage.

So, in our case, the comparator/Timer IC gives a low output, which does not allow the current to flow through the transistor and keeps the switch in NC position. But the NC pin is connected to the ground as shown in the figure thus, the pump is off. Similarly, the comparator/Timer IC gives a high output that turns on the transistor, and the relay coil is magnetized, thus moving the switch to NO position. This pin is connected to the pump, which starts the pump.

Simulation Setup

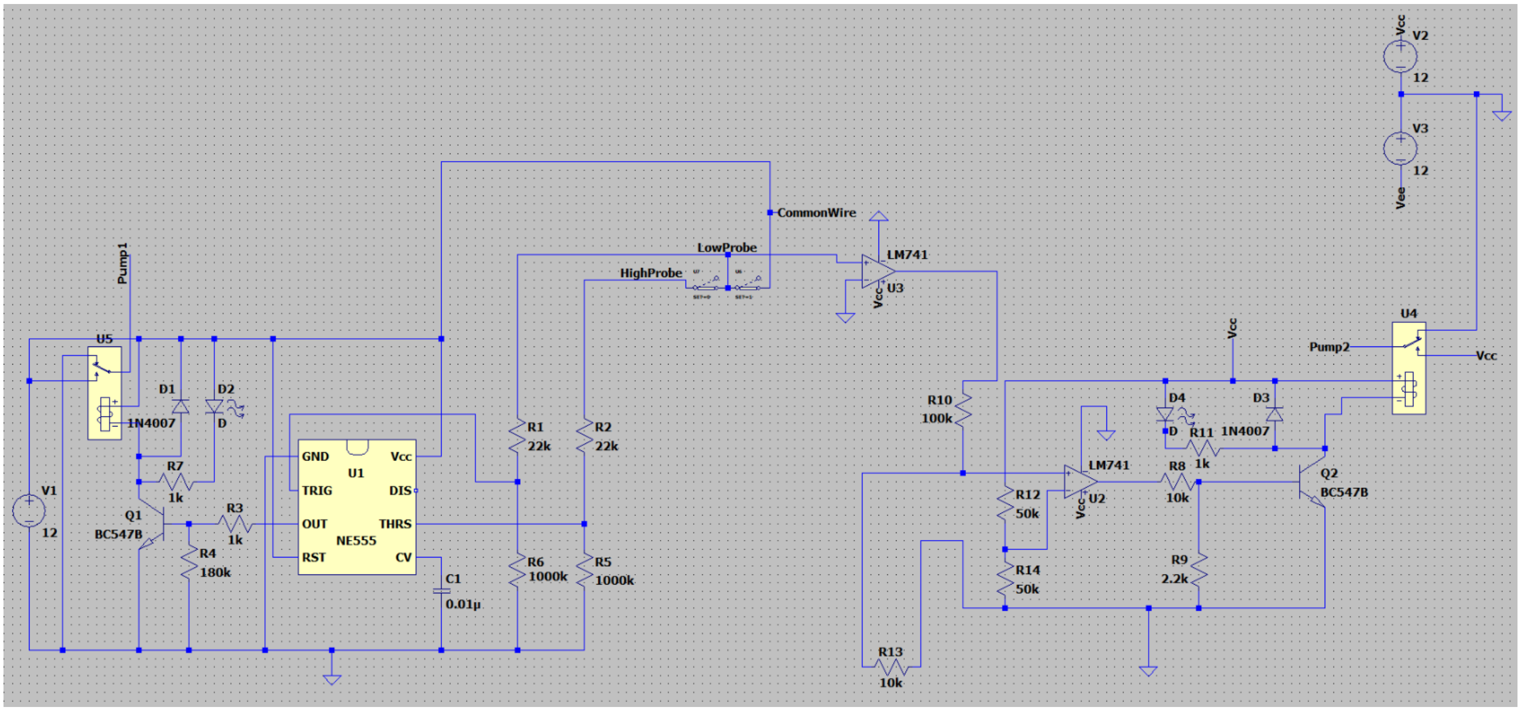


Fig 10 .Simulation Setup in LTSpice

The above figure shows the simulation setup of the Automatic Water Irrigation Circuit. The circuit corresponding to IC 555 is used for Automatic Water level sensing. The circuit at the right with LM-741 (U2) is used for soil moisture sensing. The Opamp LM-741 (U3) is used as a switch to turn ON the Moisture tester under required conditions.

For simulating the functionality of water filling in the tank, switches have been employed to simulate the working of water level probes. Due to limitations of components available in LTSpice, the pump has been replaced by a node 'Pump1' and 'Pump2'.

Water Level Monitor Setup: NE555 is used for switching the pump 'ON' or 'OFF' depending on the level of water present in the tank. The Trigger Input (Pin 2) is connected to the voltage divider between the 'Low Probe" and the common ground, while the threshold Input (Pin 5) is connected between the 'High Probe' and similar voltage divider network. The resistor values R1 = R2 = 22kΩ and R6 = R5 = 1MΩ. The resistor's values are chosen such that when the circuit contact is made with the probe, the Pin5 and Pin6 have a voltage level greater 2/3Vcc and Vcc/3, respectively. The output of the IC555 is connected to a relay driving circuit, which consists of an NPN BC547B transistor, 12 V SPDT relay, and 1N4007 diode. A LED has been added to across the relay circuit, which is turned on when the relay is Normally ON position.

Water Level Switch Setup: The Opamp (U3) is used as a comparator. The Non-Inverting terminal is connected to the "Low Probe." The Inverting terminal is grounded, due to which the comparator output will be high only if the Non-inverting terminal has a voltage drop greater than zero.

Moisture Tester Setup: The Opamp (U2) is used for switching the Irrigation Pump. The Non-Inverting terminal is connected to a voltage divider between resistor R10 = (100k) and the two probes inserted in the soil. To simulate the working of the soil, a resistor R13 has been placed. The inverting terminal of the Opamp is connected to a potentiometer which can be tuned to detect the required moisture content of the soil. For simulation, the potentiometer is placed at the middle position. When the moisture content is low, the comparator outputs a high signal switch on the relay circuit, consisting of an NPN BC547B transistor, 12 V SPDT relay, and 1N4007 diode. A LED has been added to across the relay circuit, which is turned on when the relay is Normally ON position.

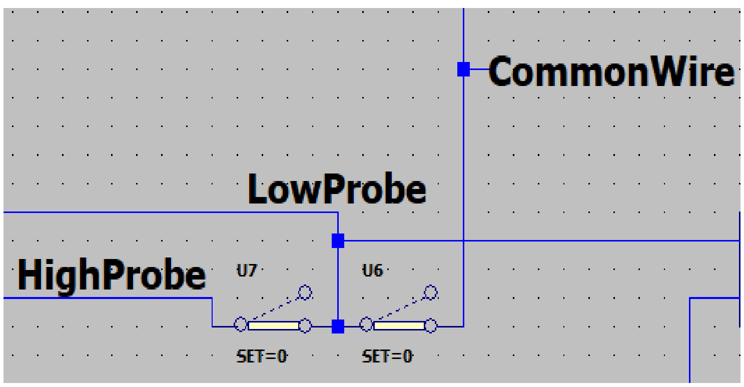
Simulation Results

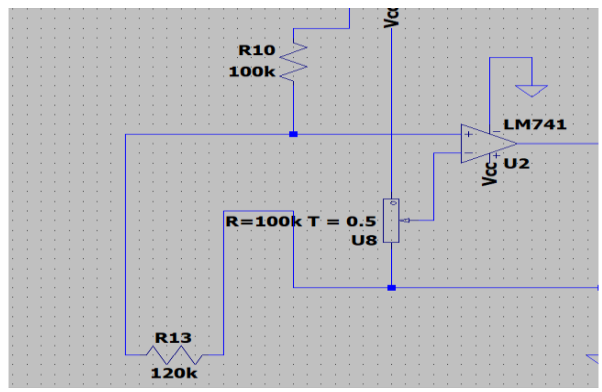
There are different possibilities under which pump-1 or pump-2 may be turned on. Four possibilities of such cases have been discussed, and the result is verified by simulating the setup on LTSpice. For demonstrating the action of an ON pump, the supply voltage to the pump is monitored. When the pump is turned on, the supply voltage to the pump is 12 V. In the case when the pump is turned off, the supply voltage to the pump is 0V.

Case 1: The soil is dry, and the water tank is empty.

In this case, the soil is dry, and the tank is empty. So initially, no water is being supplied to the soil. Then the pump associated with the water tank gets triggered by the relay, and the water level starts rising; simultaneously, water is being supplied to the soil.

An empty tank is indicated by opening both switches between the common wire and low probe, high probe. A high resistance value of 120kΩ represents dry soil.





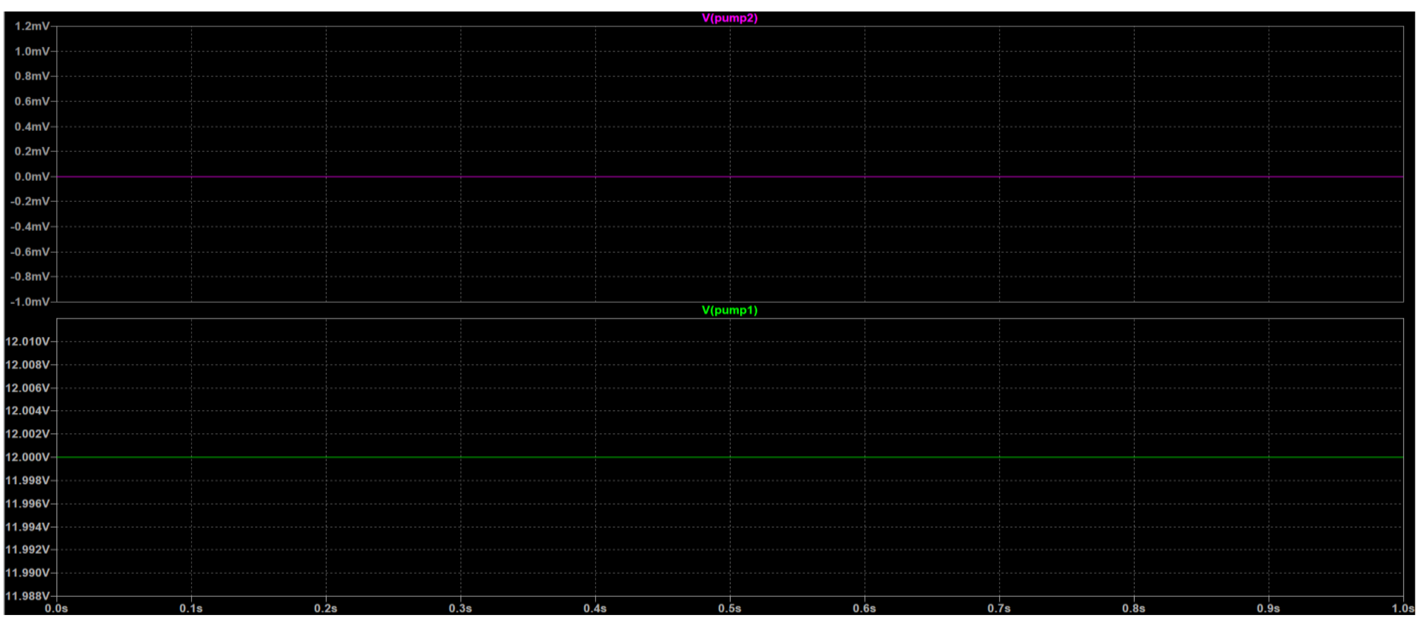


Fig 11.Simulation Result: Case-1

Case 2: The soil is dry, and the water tank is full.

In this case, since the soil is dry and the tank is full; therefore the pump associated with the soil moisture unit is triggered by the relay, and water is being supplied to the soil.

A filled tank is indicated by closing both switches between the common wire and low probe, high probe. Dry soil is represented by a high resistance value of 120kΩ.

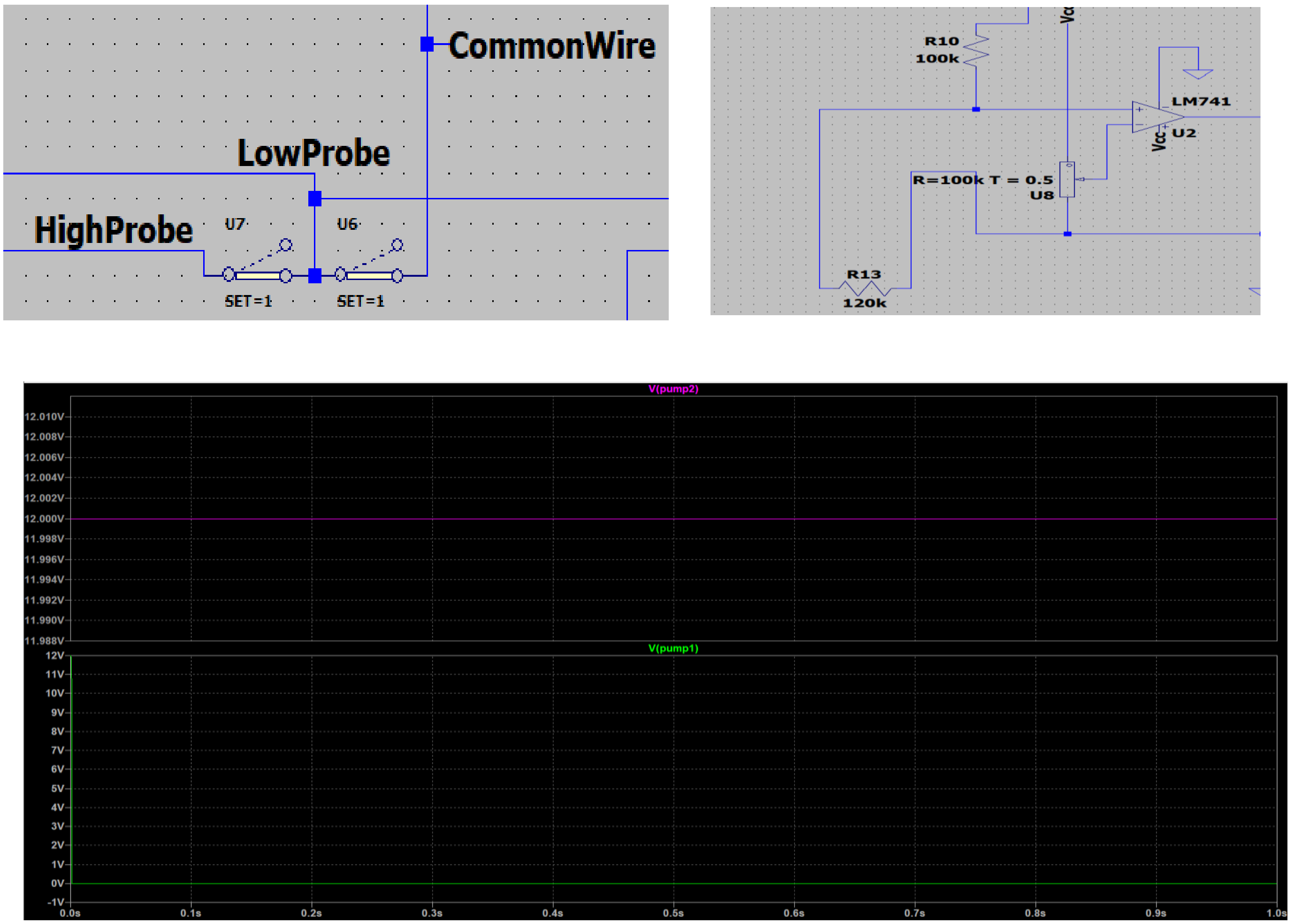


Fig 12.Simulation Result: Case-2

Case 3: The soil is wet, and the water tank is empty.

In this case, the output of the U2 Opamp is low, which switches off the pump2, but the tank is filled by switching on the pump1.

An empty Bucket is indicated by opening both switches between the common wire and low probe, high probe. Wet soil is represented by a high resistance value of 10kΩ.

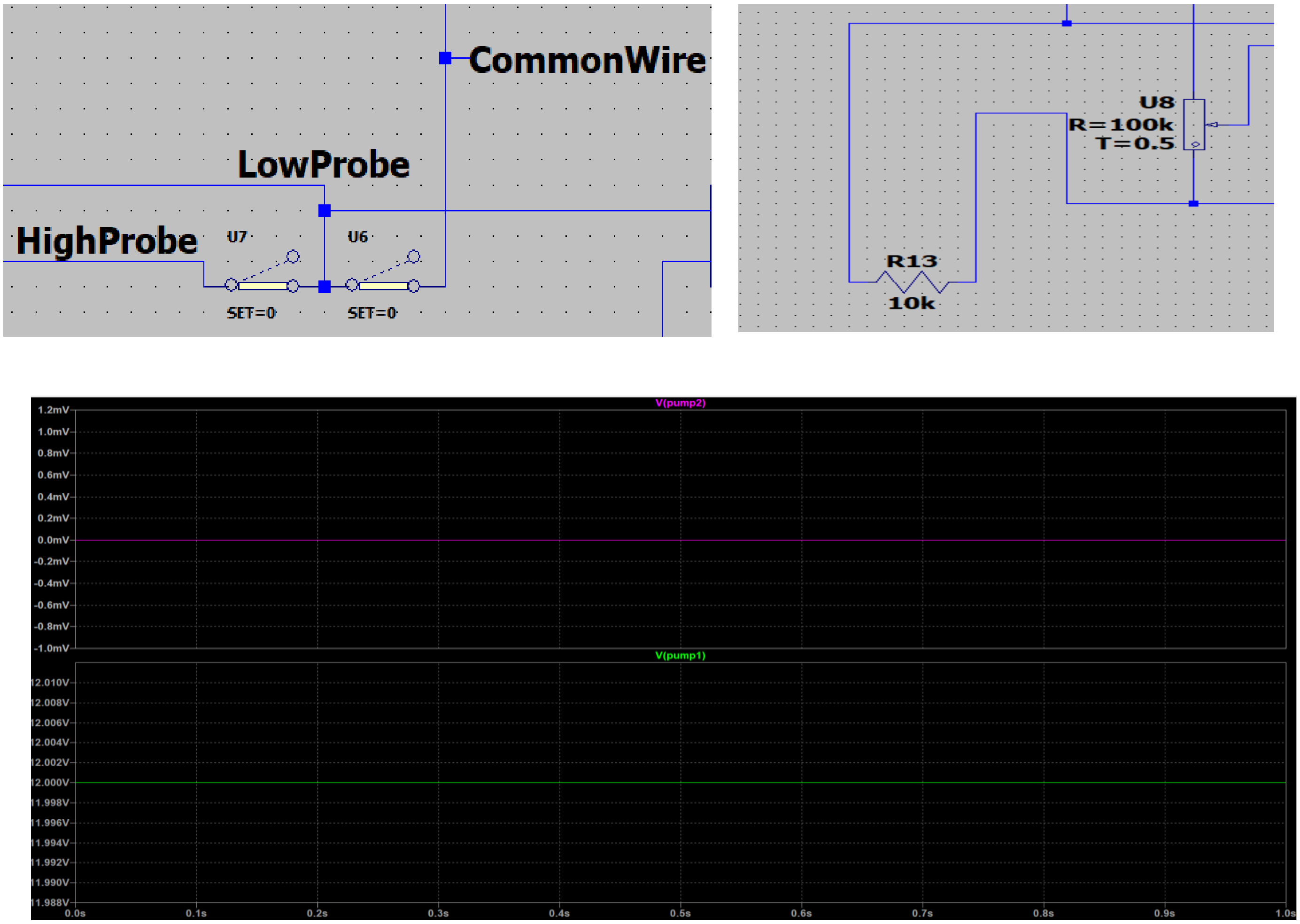
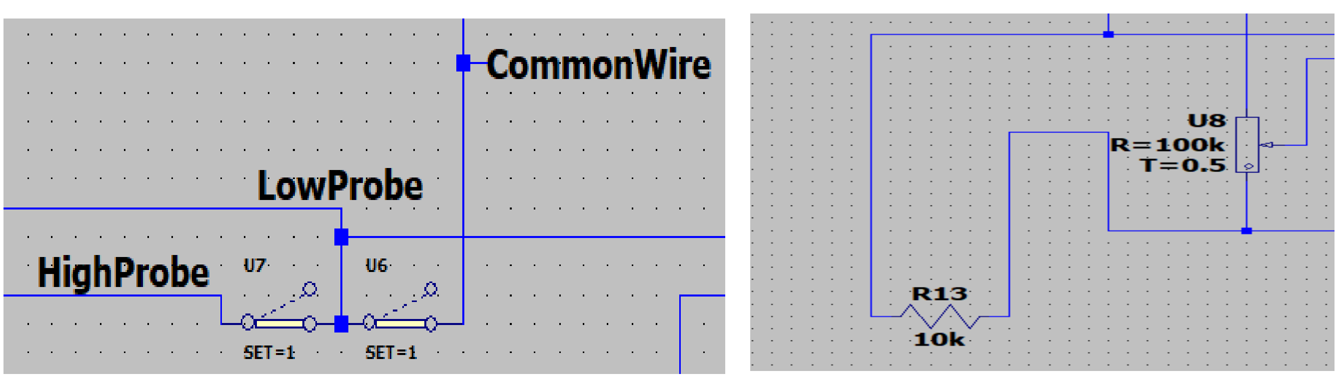


Fig 13.Simulation Result: Case-3

Case 4: The soil is wet, and the water tank is full.

Since all the conditions are met, so the whole circuit is in the off state, i.e., both the pumps are off.

A filled tank is indicated by closing both switches between the common wire and low probe, high probe. Wet soil is represented by a high resistance value of 10kΩ.



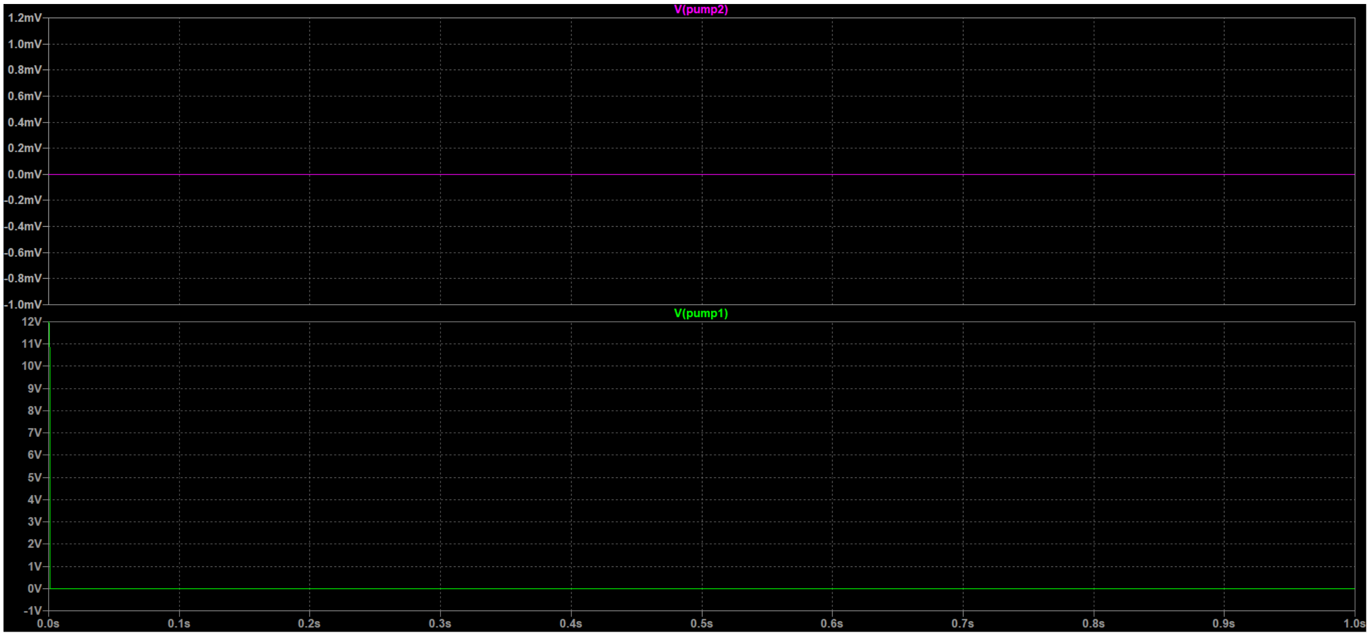


Fig 14.Simulation Result: Case-4